SBR-Matched Dynamics Settings: the Six Best Settings per Mode and Target

Generated by `analysis/write_sbr_settings_tex.py`

1 How these were found

- **Metric:** Johnson & Healy (2024) speech-to-background ratio: speech-active RMS of processed Dialogue over the RMS of the processed sum of Music, Background, Safety and Other (broadband, unweighted, both channels).
- **Targets:** HI +5.1 dB (J&H 95 % window 4.7–5.8 dB); NH +12.2 dB (window 2.8–22.4 dB).
- **One setting for all three scenes:** the objective is $J = \max_{\text{scene}} |\text{SBR} - \text{target}| + 0.02\,\bar{P} +$ penalty, where $\bar{P}$ is the mean gain reduction in dB (prefers gentler processing) and the penalty enforces Dialogue compression of 1.5–6 dB mean in every scene (group compression and Combined).
- **Dynamics only:** key gain and every makeup gain fixed at 0 dB.
- **Search:** Latin Hypercube (800 / 1500 / 2000 points), then an elitist local search per target.
- **Columns #1–#6:** the best *distinct* settings in order of J. Each differs from every setting to its left by an RMS distance of at least 0.10 (parameters scaled to their search ranges), so the columns show different ways of reaching the target, not near-copies of #1. #1 is the setting used for the presets, renders and Unity handoff (those use it rounded to slider precision).
- **Chosen for the Unity test sequence** (column marked **Unity**): Basic unmasking HI #1 and NH #3; Group compression HI #1 and NH #1. Combined (group compression, then Basic unmasking keyed on uncompressed Dialogue) is not chosen yet; its #1 is marked provisional.
- Scene SBRs unprocessed: Construction +1.31, Public Transit −4.50, Restaurant −2.71 dB; raw Dialogue dynamic range (99th−30th percentile of 125 ms levels) 14.3, 14.7, 14.5 dB.

2 Unity settings: SBR +5.0 dB on Construction

One setting per Unity mixer snapshot, each searched so the Construction scene lands at $+5.0 \pm 0.10$ dB (objective: error beyond ± 0.05 dB penalized, then least gain reduction). Key and makeup gains 0 dB; Dialogue compressed 1.5–6 dB in the two compression modes. Values are the rounded ones shipped to Unity; the SBR rows were re-measured with those rounded values. Public Transit and Restaurant are for reference only.

Table 1: Web-engine parameters of the three Unity settings (— = not part of that mode).

Parameter		**Unmasked**	**GroupCompress**	**GroupCompressed>Unmasked**
Dialogue comp. threshold	dB	—	-27.8	-38.9
Dialogue comp. ratio	:1	—	3.60	2.28
Dialogue comp. attack	ms	—	1.4	17.1
Dialogue comp. release	ms	—	170	85
Music comp. threshold	dB	—	-35.6	-33.9
Music comp. ratio	:1	—	7.25	6.34
Background comp. threshold	dB	—	-58.2	-37.5
Background comp. ratio	:1	—	1.73	8.75
Safety comp. threshold	dB	—	-53.6	-45.0
Safety comp. ratio	:1	—	3.30	5.35
Other comp. threshold	dB	—	-54.3	-38.4
Other comp. ratio	:1	—	5.18	3.80
Comp. knee (all groups)	dB	—	19.7	22.0
Comp. attack (4 background groups)	ms	—	6.8	20.6
Comp. release (4 background groups)	ms	—	363	106
Duck threshold	dB	-42.4	—	-61.4
Duck ratio	:1	13.41	—	6.07
Duck knee	dB	7.5	—	7.4
Duck attack	ms	9.0	—	38.9
Duck release	ms	433	—	22
Duck max reduction	dB	4.0	—	8.1
Key / makeup gain	dB	0	0	0
SBR, Constr. (reference)	dB	**+4.98**	**+5.00**	**+5.00**
SBR, Transit	dB	-0.85	+2.37	+3.20
SBR, Rest.	dB	+0.93	+2.64	+3.53
SBR, Constr., LL Dynamics values (emulated, 1024-sample lag)	dB	+5.00	+5.04	+5.00

Table 2: LL Dynamics values to enter in Unity (Construction web-test levels; add each group's level offset Δ in Unity). Mean GR = speech-active mean gain reduction on Construction.

Snapshot	Effect	Group	Thr. dB	Ratio	Knee dB	Atk. ms	Rel. ms	Max GR dB	Mean GR dB
Unmasked	Sidechain	Music, Bkg., FX	-34.7	13.41	7.5	9.0	433	4.0	3.8
GroupCompress	Compressor	Dialogue	-24.8	3.60	19.7	1.4	170	60.0	2.2
GroupCompress	Compressor	Music	-27.4	7.25	19.7	6.8	363	60.0	0.6
GroupCompress	Compressor	Background	-53.5	1.73	19.7	6.8	363	60.0	10.4
GroupCompress	Compressor	Safety	-46.0	3.30	19.7	6.8	363	60.0	1.1
GroupCompress	Compressor	Other	-46.7	5.18	19.7	6.8	363	60.0	11.5
GroupCompressed>Unmasked	Compressor	Dialogue	-34.5	2.28	22.0	17.1	85	60.0	2.6
GroupCompressed>Unmasked	Compressor	Music	-26.6	6.34	22.0	20.6	106	60.0	0.2
GroupCompressed>Unmasked	Compressor	Background	-33.1	8.75	22.0	20.6	106	60.0	2.9
GroupCompressed>Unmasked	Compressor	Safety	-39.0	5.35	22.0	20.6	106	60.0	0.4
GroupCompressed>Unmasked	Compressor	Other	-32.3	3.80	22.0	20.6	106	60.0	0.9
GroupCompressed>Unmasked	Sidechain	Music, Bkg., FX	-52.5	6.07	7.4	38.9	22	8.1	6.5

3 Basic unmasking

Basic unmasking: HI target (+5.1 dB)

Parameter		#1 (Unity)	#2	#3	#4	#5	#6
Duck threshold	dB	-43.6	-55.5	-66.3	-42.0	-43.9	-53.0
Duck ratio	:1	18.83	1.71	4.37	15.14	18.24	1.50
Duck knee	dB	20.3	11.6	6.8	16.8	6.9	0.0
Duck attack	ms	31.9	47.3	9.5	48.0	37.7	19.0
Duck release	ms	607	609	500	701	459	538
Duck max reduction	dB	8.0	7.3	6.7	22.8	24.1	26.9
Key (Dialogue) gain	dB	0	0	0	0	0	0
SBR, Constr.	dB	+7.85	+7.89	+8.00	+7.79	+7.94	+7.99
SBR, Transit	dB	+2.35	+2.31	+2.19	+2.20	+2.18	+2.14
SBR, Rest.	dB	+4.16	+4.26	+4.00	+4.18	+4.27	+3.99
Worst-scene |SBR−target|	dB	2.75	2.79	2.91	2.90	2.92	2.96
Mean duck, avg. of scenes	dB	-6.3	-6.2	-6.3	-7.2	-7.9	-6.2
Duck engaged, avg. of scenes	%	91	94	97	90	85	90
Search objective J		2.88	2.92	3.04	3.04	3.08	3.09

Basic unmasking: NH target (+12.2 dB)

Parameter		#1	#2	#3 (Unity)	#4	#5	#6
Duck threshold	dB	-55.4	-58.5	-59.6	-50.3	-55.3	-48.6
Duck ratio	:1	9.77	5.83	9.34	15.68	18.07	18.18
Duck knee	dB	0.0	13.1	21.1	6.9	16.3	7.6
Duck attack	ms	47.1	35.2	42.5	37.0	26.9	30.1
Duck release	ms	517	392	402	874	339	989
Duck max reduction	dB	26.2	17.2	15.9	29.0	26.5	24.7
Key (Dialogue) gain	dB	0	0	0	0	0	0
SBR, Constr.	dB	+14.53	+14.67	+14.76	+14.75	+14.73	+14.72
SBR, Transit	dB	+9.86	+9.72	+9.66	+9.72	+9.77	+9.56
SBR, Rest.	dB	+13.74	+13.22	+12.76	+12.23	+14.08	+11.64
Worst-scene |SBR−target|	dB	2.34	2.48	2.56	2.55	2.53	2.64
Mean duck, avg. of scenes	dB	-16.2	-14.2	-13.7	-15.3	-16.9	-14.9
Duck engaged, avg. of scenes	%	91	94	96	94	92	94
Search objective J		2.67	2.77	2.83	2.86	2.86	2.93

4 Group compression

Group compression: HI target (+5.1 dB)

Parameter		#1 (Unity)	#2	#3	#4	#5	#6
Dialogue comp. threshold	dB	-38.9	-41.2	-39.4	-42.7	-43.8	-33.1
Dialogue comp. ratio	:1	3.49	2.57	3.62	2.59	2.29	4.00
Dialogue comp. attack	ms	9.0	2.0	13.2	12.8	13.7	7.6
Dialogue comp. release	ms	170	30	234	133	88	294
Music comp. threshold	dB	-44.8	-44.9	-44.7	-48.9	-49.7	-48.2
Music comp. ratio	:1	6.81	7.73	7.15	7.03	8.66	5.65
Background comp. threshold	dB	-53.2	-55.3	-57.9	-45.5	-43.0	-58.0
Background comp. ratio	:1	4.66	8.47	5.65	8.77	8.56	3.95
Safety comp. threshold	dB	-57.7	-64.9	-62.8	-46.0	-60.9	-57.2
Safety comp. ratio	:1	7.62	10.00	7.22	7.45	8.30	3.37
Other comp. threshold	dB	-59.6	-52.8	-59.1	-50.0	-54.6	-53.7
Other comp. ratio	:1	7.39	4.51	5.90	7.70	10.00	4.25
Comp. knee (all groups)	dB	5.1	14.4	2.1	20.7	20.2	10.7
Comp. attack (4 background groups)	ms	19.3	23.0	19.9	8.1	9.5	16.1
Comp. release (4 background groups)	ms	461	821	391	555	568	46
Makeup gain (all groups)	dB	0	0	0	0	0	0
SBR, Constr.	dB	+5.55	+4.95	+5.87	+5.94	+6.04	+5.71
SBR, Transit	dB	+5.26	+5.79	+5.86	+5.28	+5.15	+5.33
SBR, Rest.	dB	+4.61	+4.37	+4.77	+4.15	+4.39	+4.04
Worst-scene |SBR−target|	dB	0.49	0.73	0.77	0.95	0.94	1.06
Dialogue comp. GR, avg.	dB	-3.7	-3.6	-4.0	-4.3	-3.7	-2.4
Background comp. GR, avg.	dB	-10.3	-11.2	-11.3	-9.1	-10.4	-7.2
Dialogue dyn. range, avg.	dB	8.4	6.7	9.6	9.2	9.1	9.2
Search objective J		0.77	1.03	1.07	1.22	1.23	1.25

Group compression: NH target (+12.2 dB)

Parameter		#1 (Unity)	#2	#3	#4	#5	#6
Dialogue comp. threshold	dB	-31.8	-34.1	-31.0	-39.0	-35.9	-39.3
Dialogue comp. ratio	:1	3.47	3.33	3.39	2.00	2.42	2.67
Dialogue comp. attack	ms	5.7	9.3	2.0	17.7	8.7	7.2
Dialogue comp. release	ms	284	274	274	260	94	139
Music comp. threshold	dB	-48.5	-50.5	-56.5	-56.5	-65.0	-61.6
Music comp. ratio	:1	8.41	4.98	9.07	7.80	6.34	6.36
Background comp. threshold	dB	-56.4	-54.9	-51.6	-61.8	-51.3	-55.9
Background comp. ratio	:1	5.33	5.21	4.57	2.91	5.74	5.75
Safety comp. threshold	dB	-59.3	-59.7	-51.0	-49.8	-57.3	-56.8
Safety comp. ratio	:1	4.17	4.68	4.98	5.50	7.22	8.85
Other comp. threshold	dB	-60.1	-62.7	-61.9	-54.0	-59.5	-61.9
Other comp. ratio	:1	9.32	8.70	9.28	6.79	2.98	3.92
Comp. knee (all groups)	dB	6.6	4.4	10.4	17.9	14.4	18.7
Comp. attack (4 background groups)	ms	10.7	10.2	5.7	20.7	31.5	38.3
Comp. release (4 background groups)	ms	538	595	517	669	995	1000
Makeup gain (all groups)	dB	0	0	0	0	0	0
SBR, Constr.	dB	+12.47	+12.84	+13.09	+13.12	+13.21	+12.98
SBR, Transit	dB	+12.41	+12.25	+11.30	+12.29	+12.97	+12.71
SBR, Rest.	dB	+11.92	+11.73	+11.89	+11.21	+11.20	+11.11
Worst-scene |SBR−target|	dB	0.28	0.64	0.90	0.99	1.01	1.09
Dialogue comp. GR, avg.	dB	-1.9	-2.2	-2.4	-2.6	-1.7	-3.6
Background comp. GR, avg.	dB	-13.1	-13.6	-14.3	-12.6	-13.9	-14.7
Dialogue dyn. range, avg.	dB	8.9	9.3	7.6	11.0	9.1	8.4
Search objective J		0.58	0.95	1.23	1.30	1.35	1.46

5 Combined

Combined: HI target (+5.1 dB)

Parameter		#1 (Unity, prov.)	#2	#3	#4	#5	#6
Dialogue comp. threshold	dB	-44.7	-44.1	-39.1	-39.8	-40.2	-39.0
Dialogue comp. ratio	:1	2.76	2.56	2.83	2.53	3.51	3.60
Dialogue comp. attack	ms	6.2	6.8	19.5	17.2	19.8	18.1
Dialogue comp. release	ms	48	69	205	142	223	159
Music comp. threshold	dB	-62.4	-65.0	-62.4	-64.4	-52.9	-43.8
Music comp. ratio	:1	4.88	4.60	4.26	3.37	9.70	8.43
Background comp. threshold	dB	-58.0	-49.9	-36.2	-42.7	-47.8	-51.7
Background comp. ratio	:1	3.80	5.70	10.00	9.72	8.08	8.73
Safety comp. threshold	dB	-30.2	-27.8	-48.3	-43.9	-27.2	-58.0
Safety comp. ratio	:1	3.18	4.74	8.98	9.55	4.02	5.32
Other comp. threshold	dB	-29.3	-27.9	-42.7	-45.5	-20.0	-64.6
Other comp. ratio	:1	4.78	6.07	7.15	6.93	6.64	3.29
Comp. knee (all groups)	dB	16.6	17.5	17.4	18.1	14.6	3.4
Comp. attack (4 background groups)	ms	45.4	46.2	37.3	38.8	48.6	30.6
Comp. release (4 background groups)	ms	20	112	399	344	927	302
Duck threshold	dB	-59.1	-57.2	-43.1	-38.1	-58.6	-25.5
Duck ratio	:1	7.04	6.71	15.72	19.23	7.94	7.83
Duck knee	dB	5.7	3.9	10.7	13.5	6.4	7.1
Duck attack	ms	47.2	38.5	32.5	29.2	21.1	48.1
Duck release	ms	152	116	378	249	20	772
Duck max reduction	dB	13.6	14.6	28.3	28.6	11.8	21.0
Key (Dialogue) gain	dB	0	0	0	0	0	0
Makeup gain (all groups)	dB	0	0	0	0	0	0
SBR, Constr.	dB	+5.20	+5.29	+6.00	+6.07	+5.98	+5.65
SBR, Transit	dB	+5.11	+5.27	+4.70	+5.87	+5.92	+5.05
SBR, Rest.	dB	+5.05	+4.88	+4.36	+4.12	+4.19	+3.97
Worst-scene |SBR−target|	dB	0.10	0.22	0.90	0.98	0.91	1.13
Mean duck, avg. of scenes	dB	-10.7	-10.5	-7.1	-3.3	-7.9	-0.0
Duck engaged, avg. of scenes	%	87	85	85	77	82	11
Dialogue comp. GR, avg.	dB	-4.7	-4.6	-3.0	-2.8	-3.8	-2.8
Background comp. GR, avg.	dB	-5.4	-6.9	-6.6	-7.6	-6.9	-9.1
Dialogue dyn. range, avg.	dB	6.9	7.5	10.4	10.0	10.4	9.8
Search objective J		0.52	0.66	1.24	1.25	1.28	1.37

Combined: NH target (+12.2 dB)

Parameter		#1 (Unity, prov.)	#2	#3	#4	#5	#6
Dialogue comp. threshold	dB	-33.6	-36.1	-41.4	-36.5	-39.9	-40.4
Dialogue comp. ratio	:1	3.98	3.88	3.32	3.08	3.29	2.32
Dialogue comp. attack	ms	11.1	10.1	16.1	15.7	14.6	15.8
Dialogue comp. release	ms	198	146	250	250	199	80
Music comp. threshold	dB	-22.0	-20.5	-57.5	-61.2	-62.3	-63.8
Music comp. ratio	:1	5.68	4.17	5.37	3.21	3.32	2.04
Background comp. threshold	dB	-43.6	-53.8	-55.2	-52.1	-54.5	-44.4
Background comp. ratio	:1	9.98	10.00	4.29	5.39	3.87	8.80
Safety comp. threshold	dB	-49.6	-51.8	-65.0	-51.3	-61.3	-43.1
Safety comp. ratio	:1	3.19	4.30	8.75	9.83	10.00	8.86
Other comp. threshold	dB	-21.3	-26.7	-65.0	-59.0	-65.0	-46.4
Other comp. ratio	:1	1.64	1.92	6.06	6.50	5.70	6.11
Comp. knee (all groups)	dB	10.0	8.4	19.4	14.0	18.0	19.1
Comp. attack (4 background groups)	ms	17.3	21.6	8.2	17.3	12.4	39.4
Comp. release (4 background groups)	ms	564	588	836	733	784	189
Duck threshold	dB	-60.8	-61.3	-21.0	-24.4	-26.1	-49.0
Duck ratio	:1	15.19	16.96	14.01	12.45	14.44	18.56
Duck knee	dB	18.5	22.8	2.7	7.3	4.5	12.7
Duck attack	ms	30.6	28.8	28.9	35.3	37.8	33.5
Duck release	ms	130	127	347	333	468	464
Duck max reduction	dB	19.5	22.1	6.2	5.7	10.7	28.1
Key (Dialogue) gain	dB	0	0	0	0	0	0
Makeup gain (all groups)	dB	0	0	0	0	0	0
SBR, Constr.	dB	+12.02	+11.79	+13.12	+13.32	+13.02	+12.63
SBR, Transit	dB	+12.40	+12.57	+11.88	+11.97	+11.38	+13.45
SBR, Rest.	dB	+11.99	+12.23	+11.38	+11.16	+11.11	+12.49
Worst-scene |SBR−target|	dB	0.21	0.41	0.92	1.12	1.09	1.25
Mean duck, avg. of scenes	dB	-15.0	-16.5	-0.0	-0.0	-0.0	-12.3
Duck engaged, avg. of scenes	%	92	95	0	5	7	90
Dialogue comp. GR, avg.	dB	-1.7	-2.3	-4.9	-2.5	-4.0	-2.4
Background comp. GR, avg.	dB	-3.0	-5.4	-16.3	-12.9	-15.3	-6.0
Dialogue dyn. range, avg.	dB	9.5	8.7	10.2	10.3	9.8	9.7
Search objective J		0.61	0.89	1.34	1.42	1.48	1.66

6 Crossover with the mean-margin-gain sweeps

The earlier mean-margin-gain sweeps (Basic ducking, Dialogue key, max reduction fixed at 6 dB; 150 / 50 / 100 points on Construction / Public Transit / Restaurant) were re-measured as SBR. Margins use the band those sweeps used (Dialogue 400–7000 Hz, narrowed 0.85 octave each side, 20 ms frames, speech-active gate); recomputing them from the renders reproduces the sweep CSV values exactly.

Table 3: Margin gain versus SBR within each margin sweep.

Scene	Points	Pearson r	Spearman ρ	Best margin gain	SBR of that point	Unprocessed SBR
Constr.	150	+0.986	+0.999	+6.97	+7.10	+1.31
Transit	50	+0.986	+0.997	+6.68	+1.40	-4.50
Rest.	100	+0.994	+0.999	+6.82	+3.29	-2.71

Table 4: Top-5 margin-gain settings from each sweep, and the SBR they produce.

Scene	Point	Thr. dB	Ratio	Knee dB	Atk. ms	Rel. ms	Margin gain dB	SBR dB
Constr.	#126	-58.6	7.40	5.9	3.2	215	+6.97	+7.10
Constr.	#9	-53.5	9.86	16.8	5.0	354	+6.95	+7.12
Constr.	#97	-56.2	5.50	10.7	6.4	307	+6.95	+7.10
Constr.	#146	-58.1	3.59	19.0	14.9	344	+6.94	+7.14
Constr.	#27	-57.5	3.66	6.8	15.4	382	+6.93	+7.11
Transit	#3	-56.0	5.66	23.5	8.0	312	+6.68	+1.40
Transit	#16	-53.8	7.72	11.1	21.4	309	+6.60	+1.24
Transit	#34	-59.4	2.26	4.2	38.3	270	+6.59	+1.25
Transit	#27	-56.7	4.57	16.6	46.3	293	+6.58	+1.26
Transit	#33	-57.8	7.14	0.6	22.2	142	+6.57	+1.06
Rest.	#64	-56.2	4.01	17.7	1.2	294	+6.82	+3.29
Rest.	#99	-55.6	5.29	3.2	8.7	398	+6.82	+3.28
Rest.	#16	-58.6	8.86	22.1	14.1	275	+6.82	+3.28
Rest.	#66	-57.8	1.99	6.3	4.0	202	+6.82	+3.28
Rest.	#46	-56.3	9.71	0.2	17.4	320	+6.81	+3.28

Table 5: Margin gain of the SBR-matched settings. *True* margin is measured from the separately rendered processed Dialogue and processed background; the *estimator* is the sweep's `compute_masking_margin` (mix minus dry Dialogue), valid only when Dialogue is unprocessed (Basic). Last column: best estimator margin gain found in that scene's sweep.

Setting	Scene	SBR dB	True margin gain dB	Estimator margin gain dB	Best in margin sweep dB
`basic_HI`	Constr.	+7.84	+7.38	+8.74	+6.97
`basic_HI`	Transit	+2.34	+7.41	+8.46	+6.68
`basic_HI`	Rest.	+4.15	+7.19	+8.48	+6.82
`basic_NH`	Constr.	+14.78	+15.40	+19.19	+6.97
`basic_NH`	Transit	+9.69	+15.47	+18.87	+6.68
`basic_NH`	Rest.	+12.79	+15.62	+20.45	+6.82
`comp_HI`	Constr.	+5.54	+5.65	–	+6.97
`comp_HI`	Transit	+5.25	+10.79	–	+6.68
`comp_HI`	Rest.	+4.60	+10.13	–	+6.82
`comp_NH`	Constr.	+12.50	+12.06	–	+6.97
`comp_NH`	Transit	+12.43	+17.25	–	+6.68
`comp_NH`	Rest.	+11.95	+16.35	–	+6.82
`combined_HI`	Constr.	+5.21	+11.20	–	+6.97
`combined_HI`	Transit	+5.13	+16.86	–	+6.68
`combined_HI`	Rest.	+5.05	+11.83	–	+6.82
`combined_NH`	Constr.	+12.00	+18.23	–	+6.97
`combined_NH`	Transit	+12.37	+22.99	–	+6.68
`combined_NH`	Rest.	+11.97	+17.72	–	+6.82

What the crossover shows

- **Same direction, different question.** Within each margin sweep, margin gain and SBR gain rank the settings almost identically (Spearman $\rho \geq 0.997$): anything that ducks more improves both. The best margin-gain settings are also the highest-SBR settings in those sweeps.
- **But margin gain is "more is better" and SBR has an optimum.** With max reduction capped at 6 dB, the best margin settings reach SBR +7.1 (Construction), +1.4 (Public Transit) and +3.3 dB (Restaurant): near the HI target in Construction only, and well short of the NH target everywhere. Reaching +12.2 dB needs max reduction above 6 dB, which the margin sweeps never varied.
- **The margin estimator reads high.** Because it estimates the masker as mix power minus dry-Dialogue power, it overstates the true in-band margin gain: by about 1 dB at the sweeps' best points (e.g. +6.97 vs. a true +5.96 dB; the true value cannot exceed the 6 dB cap) and by 5–7 dB for the deep NH ducks. It also cannot be used for group compression, where Dialogue itself is processed.
- **At equal SBR, margin differs by mode.** The SBR-matched settings hit the same broadband SBR, yet their in-band margin gains differ widely (e.g. HI target, Construction: group compression +5.7, Combined +11.2 dB), because margin averages per-frame dB in the speech band while SBR is a broadband energy ratio.